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B.Tech. Degree VII Semester Supplementary Examination
April 2018

ME 1701 REFRIGERATION AND AIR CONDITIONING
(2012 Scheme)

(Refrigeration data books are permitted)



Time : 3 Hours

Maximum Marks : 100

PART A(Answer **ALL** questions)

(8 × 5 = 40)

- I. (a) With a neat sketch explain the working of vortex tube.
- (b) Explain the working of a Bell-Coleman cycle and derive the expression for C.O.P.
- (c) Explain with neat sketch the vapour compression refrigeration system. Also show the cycle on T-s, p-v and p-h diagrams.
- (d) What are the criteria for selection of refrigerants in a vapour compression system?
- (e) What are the functions of a refrigeration compressor? List various types of refrigerating compressors.
- (f) Explain the thermostatic expansion valve with neat sketch.
- (g) Briefly explain central and unitary systems in air conditioning.
- (h) Define specific humidity and relative humidity.

PART B

(4 × 15 = 60)

- II. (a) Describe the working of Pulse tube refrigeration with a neat sketch. (6)
- (b) Explain with neat sketch a Bootstrap cycle of air refrigeration system with evaporative cooling. (9)

OR

- III. Following data refers to Boot strap air cycle system. (15)
 Altitude of airplane – 14 km, Ambient temperature = -55°C, Airplane speed – 1.2 Mach, Ambient pressure = 0.15 bar, Cabin pressure = 0.8 bar, Cabin temperature = 24°C, Cooling load – 50 TR, Ram recovery factor – 0.9, Isentropic efficiencies of main compressor, secondary compressor and cooling turbine are 90%, 85% and 85% respectively, Main compressor pressure ratio – 5, Secondary compressor pressure ratio – 1.3 and the temperature drops along the 1st heat exchanger, 2nd heat exchanger and evaporative cooler are 65°C, 75°C and 50°C respectively. Find (i) mass flow rate of air (ii) power required and (iii) C.O.P. of the system.

(P.T.O.)

- IV. A 10 TR refrigeration system is required for a food storage system. The evaporator temperature is -20°C , and the condenser temperature is 50°C . The working fluid is R-134a. The refrigerant is superheated by 5°C in the evaporator. The refrigerant is sub cooled by 5°C at the exit of condenser and before entering the expansion valve. The refrigerant gets compressed in a double cylinder single acting reciprocating compressor having L/D as 1.1 and 80% volumetric efficiency. The compressor runs at 1200 rpm. Calculate: (i) Mass flow rate of refrigerant in kg/min, (ii) bore and stroke of compressor in mm, and (iii) theoretical power required in kW. (15)

OR

- V. (a) Explain the working of a cascaded refrigeration system with a neat sketch. (5)
 (b) Explain with neat sketch the working of aqua – ammonia vapour absorption system. (10)

- VI. A single stage, single acting reciprocating compressor has a bore of 200 mm and a stroke of 300 mm. It receives vapour refrigerant at 1 bar and delivers it at 5.5 bar. If the compression and expansion follows the law $p v^{1.3} = \text{Constant}$ and the clearance volume is 5 percent of the stroke volume, determine : (i) The power required to drive the compressor, if it runs at 500 rpm; and (ii) The volumetric efficiency of the compressor. (15)

OR

- VII. What is an evaporator? Explain with neat sketches the following evaporators. (15)
 (i) Flooded type. (ii) Dry expansion type.

- VIII. (a) Explain summer and winter air conditioning. (8)
 (b) An air conditioning plant is required to supply 60 m^3 of air per minute at a DBT of 21°C and 55% RH. The outside air is at a DBT of 28°C and 60% RH. Determine the mass of water drained and the capacity of the cooling coil. Assume the air conditioning plant to first dehumidify and then to cool the air. (7)

OR

- IX. (a) $30 \text{ m}^3/\text{min}$ of a stream of moist air at 15°C DBT and 13°C WBT are mixed with $12 \text{ m}^3/\text{min}$ of a second stream at 25°C DBT and 18°C WBT. Barometric pressure is one standard atmosphere. Determine the dry bulb and wet bulb temperatures of the resulting mixture. (10)
 (b) Briefly describe comfort chart used in air conditioning. (5)
